

# Analysis Documentation

## Student Performance Prediction — a guided statistical study

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**Executive summary.** This document investigates the Student Performance Prediction dataset (10,000 records, 8 variables) through five guided questions — describing the data, testing group differences, measuring relationships, building a predictive model, and synthesising what it all means for understanding student performance.

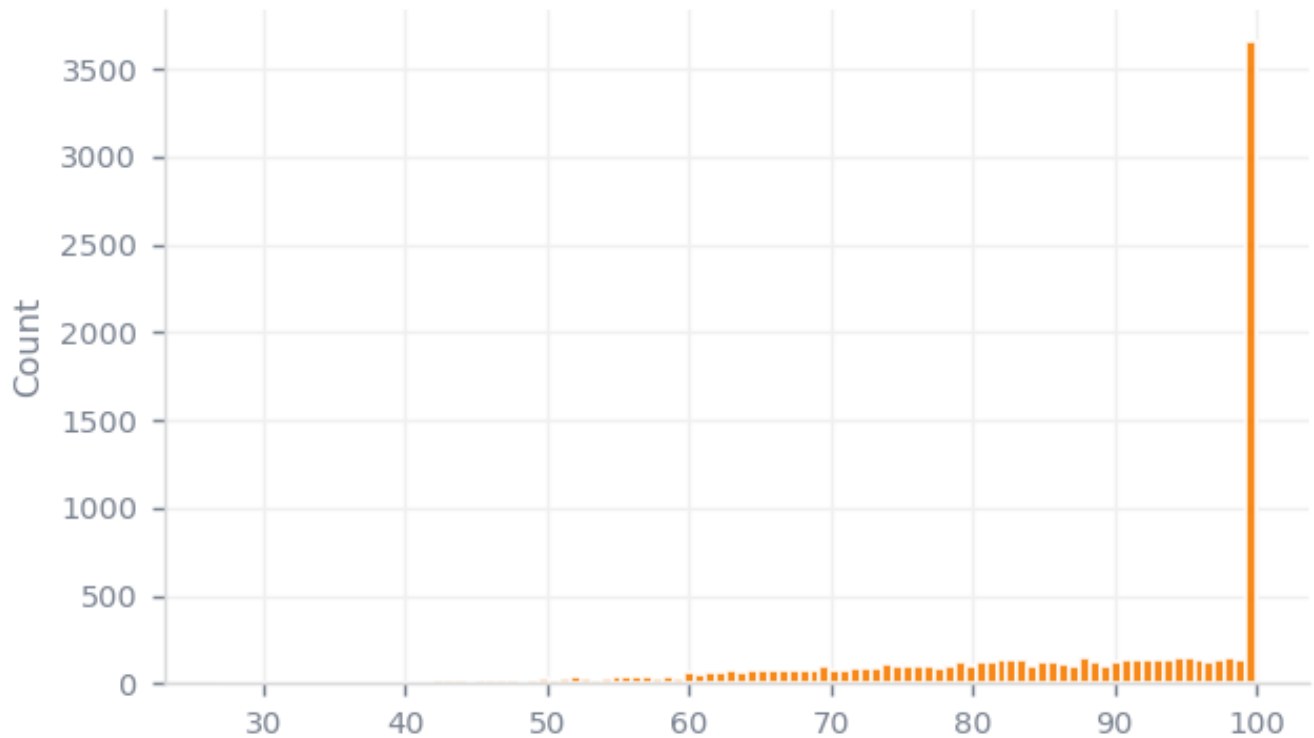
### 1. A. Descriptive Statistics — What do typical students look like?

|          | study_hours | attendance | sleep_hours | internet_usage | assignments_completed | previous_exam_score | exam_score |
|----------|-------------|------------|-------------|----------------|-----------------------|---------------------|------------|
| N        | 10000       | 10000      | 10000       | 10000          | 10000                 | 10000               | 10000      |
| Mean     | 5.9896      | 69.8846    | 6.4985      | 6.0626         | 9.9884                | 64.911              | 86.7042    |
| Median   | 6.0         | 70.0       | 6.5         | 6.0            | 10.0                  | 65.0                | 92.12      |
| Mode     | 10          | 56         | 5           | 5              | 5                     | 57                  | 100.0      |
| Variance | 10.0083     | 310.3421   | 2.9219      | 9.8481         | 36.4109               | 306.3557            | 226.7549   |
| Std      | 3.1636      | 17.6165    | 1.7094      | 3.1382         | 6.0341                | 17.503              | 15.0584    |
| Min      | 1           | 40         | 4           | 1              | 0                     | 35                  | 26.67      |
| Max      | 11          | 100        | 9           | 11             | 20                    | 95                  | 100.0      |
| Range    | 10          | 60         | 5           | 10             | 20                    | 60                  | 73.33      |
| Q1       | 3.0         | 55.0       | 5.0         | 3.0            | 5.0                   | 50.0                | 76.7275    |
| Q3       | 9.0         | 85.0       | 8.0         | 9.0            | 15.0                  | 80.0                | 100.0      |
| IQR      | 6.0         | 30.0       | 3.0         | 6.0            | 10.0                  | 30.0                | 23.2725    |
| Skewness | -0.0072     | 0.0178     | 0.0051      | -0.0135        | 0.0178                | 0.0004              | -1.0086    |
| Kurtosis | -1.2218     | -1.2075    | -1.2714     | -1.2008        | -1.1959               | -1.1856             | 0.1475     |
| Missing  | 0           | 0          | 0           | 0              | 0                     | 0                   | 0          |

Problem statement: What are the central tendencies and spread of the measured study habits and outcomes? exam\_score averages 86.70 (median 92.12, SD 15.06), ranging from 26.67 to 100.00. The distribution is left-skewed (skewness -1.01). These baselines frame every comparison and model that follows.

### 2. A. Distribution of exam\_score

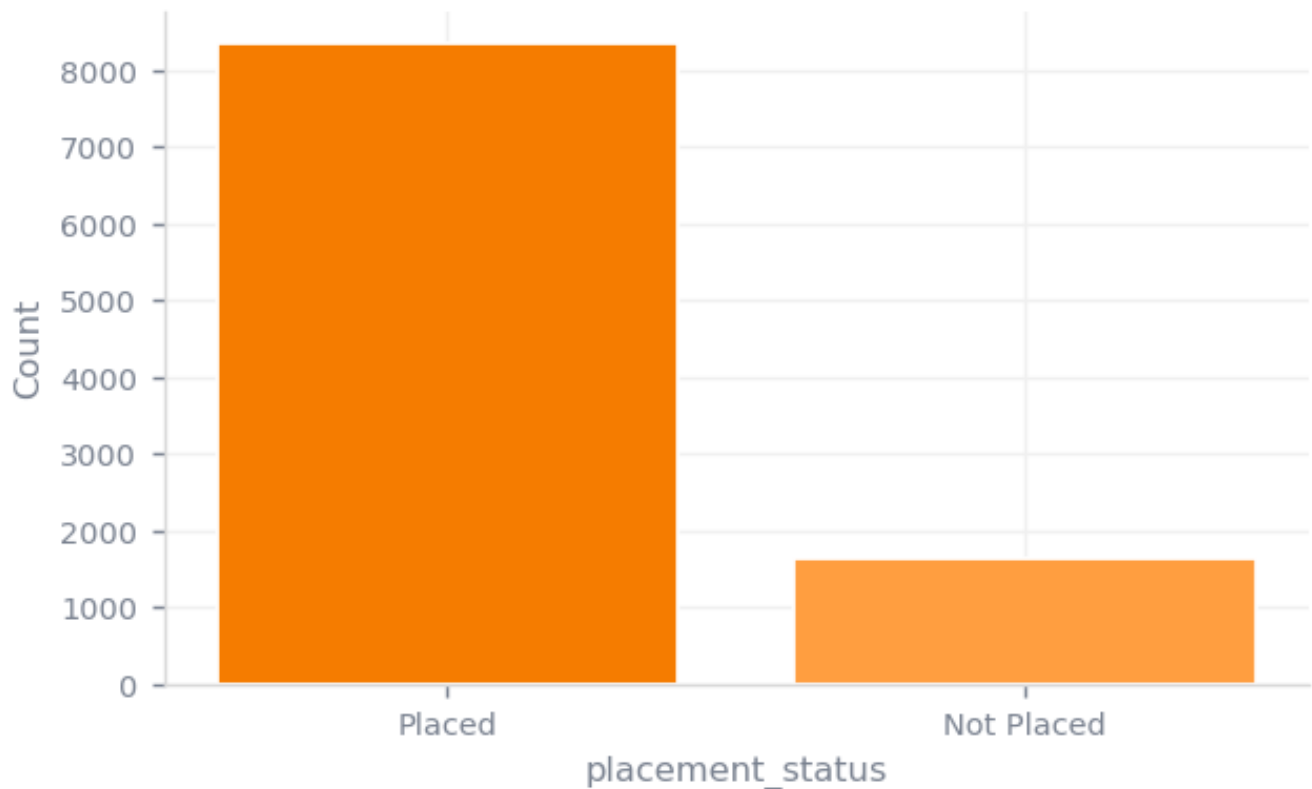
## Distribution of exam\_score



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### 3. B. Categorical Testing — Does exam\_score depend on placement\_status?

## placement\_status distribution



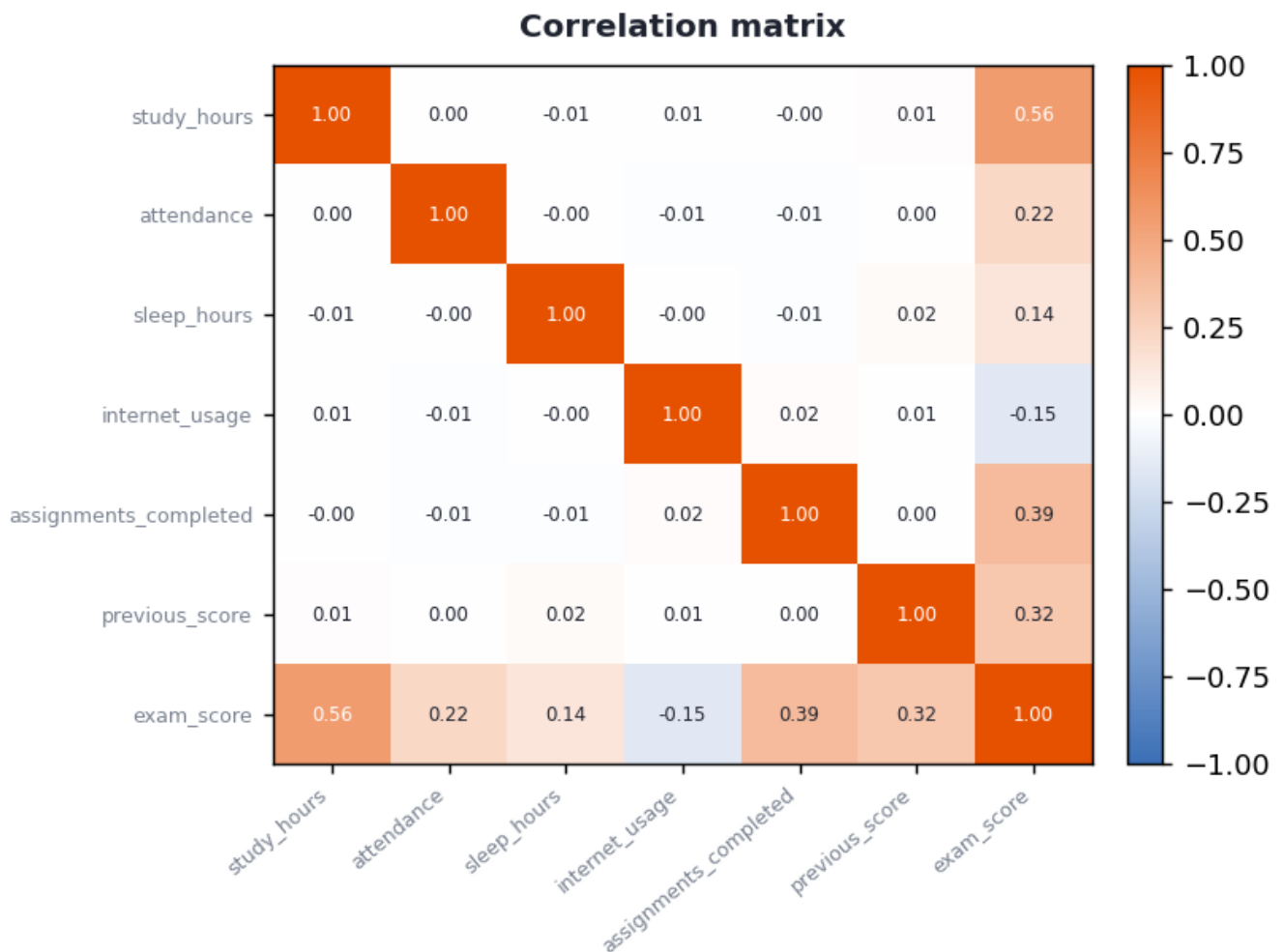
Problem statement: Are outcomes distributed evenly across placement\_status, and does exam\_score differ between its groups? placement\_status is dominated by "Placed" (83.56% of records).

### 4. B. exam\_score by placement\_status — independent t-test

|            | Mean   | SD    | N        |
|------------|--------|-------|----------|
| Placed     | 91.997 | 9.377 | 8356.000 |
| Not Placed | 59.805 | 8.147 | 1644.000 |

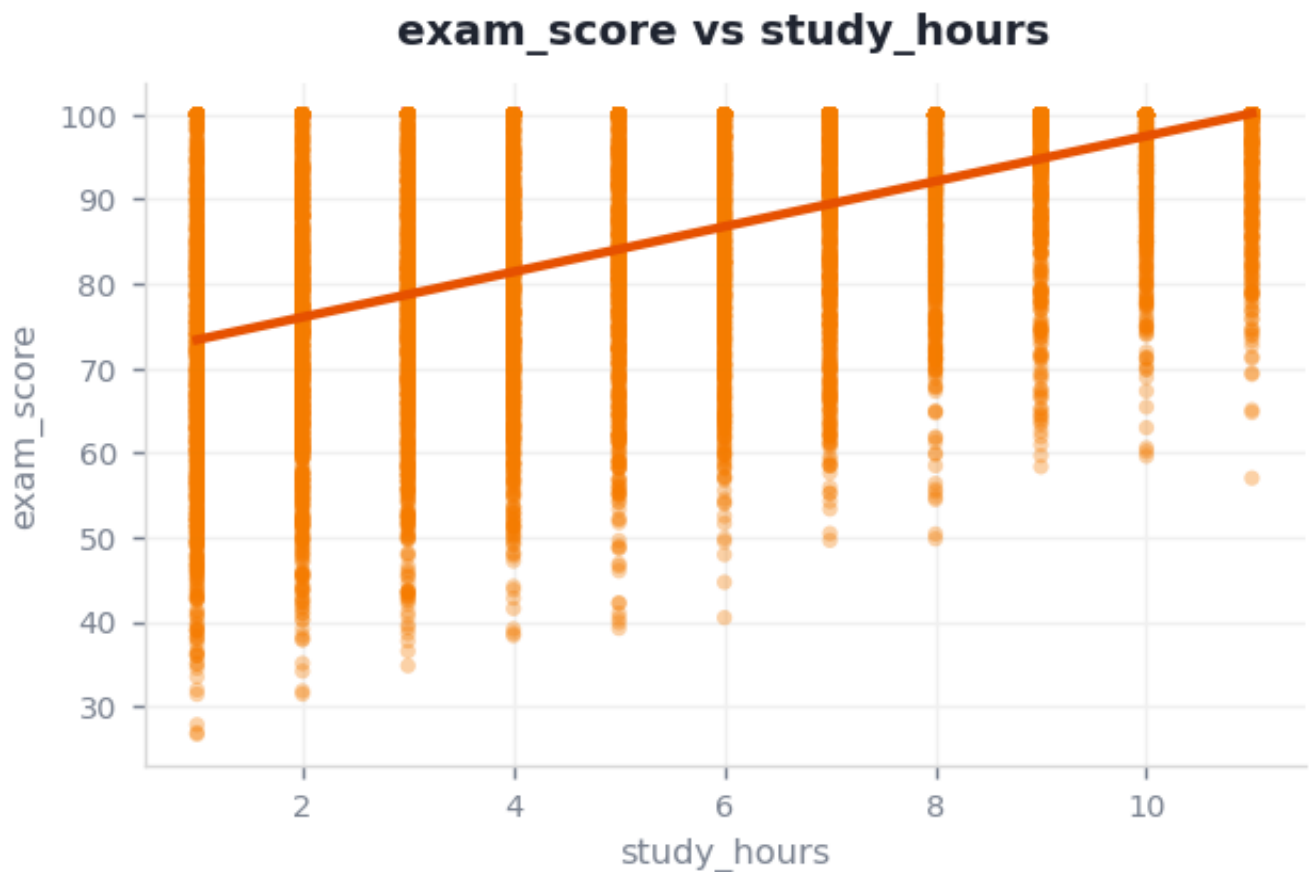
Reject  $H_0$  — exam\_score differs significantly between "Placed" and "Not Placed" ( $t = 142.69$ ,  $p < 0.001$ , Cohen's  $d = 3.50$ , large effect), so placement\_status is a meaningful lens on performance.

### 5. C. Correlation — Which factors move together?



Problem statement: Which variables are most strongly related, and could therefore drive performance? The strongest linear relationship is between study\_hours and exam\_score ( $r = 0.56$ ), a moderate positive association. Variables with high correlation may carry overlapping information.

## 6. D. Predictive Trend — Modelling exam\_score from study\_hours



Problem statement: Can exam\_score be predicted from study\_hours? A simple linear model predicts exam\_score from study\_hours as  $Y = 70.667 + 2.678 \times \text{study\_hours}$ . It explains 31.6% of the variance ( $R^2 = 0.316$ ). The slope is statistically significant ( $p < 0.001$ ), so each one-unit rise in study\_hours is associated with a change of 2.678 in exam\_score. For example, a student one standard deviation above the mean study\_hours (9.2) is predicted to score 95.2.

## 7. E. Synthesis — Putting the picture together

Tying the threads together: the descriptive profile sets expectations, categorical testing shows which groups differ, correlation highlights the factors that move with performance, and the regression turns the strongest of those into a predictive rule. Together they describe how study\_hours relate to exam\_score, give a baseline model for prediction, and flag where richer multivariate models could improve on a single predictor.